

Web Application Development

IS4102

Lecture Note 03 - Web Servers

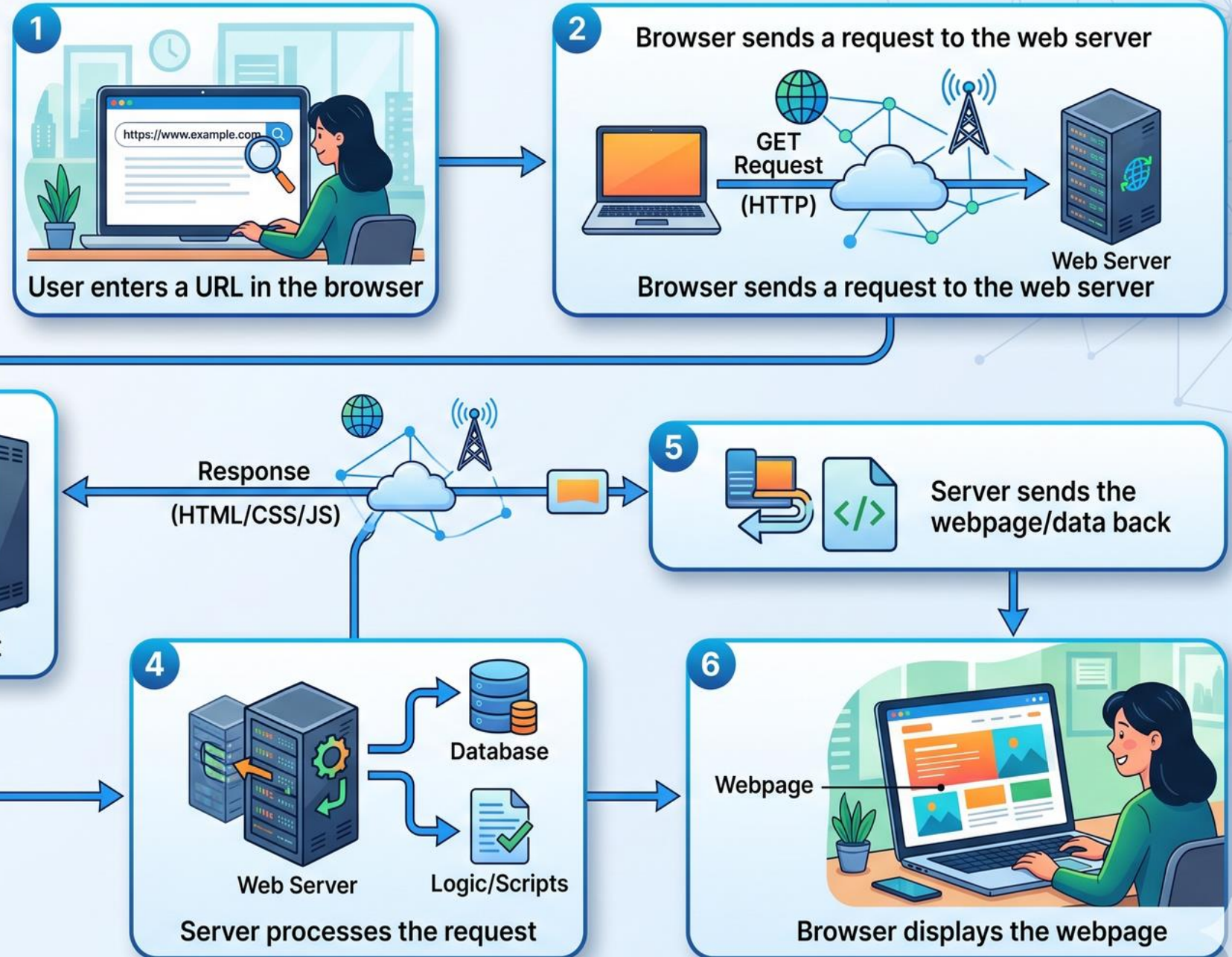
By Groups: 2, 8, 14, 20, 26

Web Server

A web server is a system that stores, processes, and delivers web pages to users through the internet or a local network.



STEP-by-STEP PROCESS: USER ENTERS URL TO WEBPAGE DISPLAY



Key Components of a Web Server

Hardware - Physical computer storing website files.

Software - Manages web requests and responses.

Network Connection - Connects the server to the internet

Database - Stores website data and user information.

Defining the Web Server

Physical Hardware

Think of it as a specialized computer. It stays on 24/7 and stores all the files (images, text, videos) that make up a website.

Specialized Software

This is the "brain" of the server. It understands web addresses (URLs) and knows how to send the right files to the right person.

Basic Architecture of a Web Server

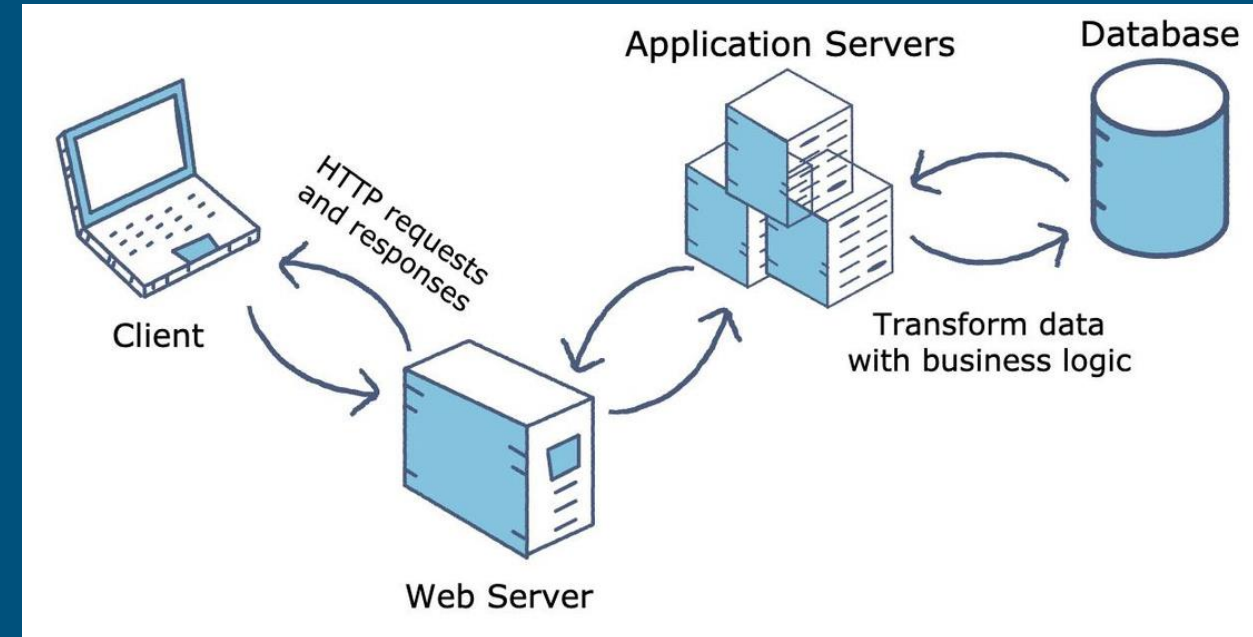
Client – User's Browser

Internet/Network–Communication Medium

Web Server – Handles Requests

Database – Stores Data

Application – Processes Business Logic



System Architecture

Client-Server Architectures

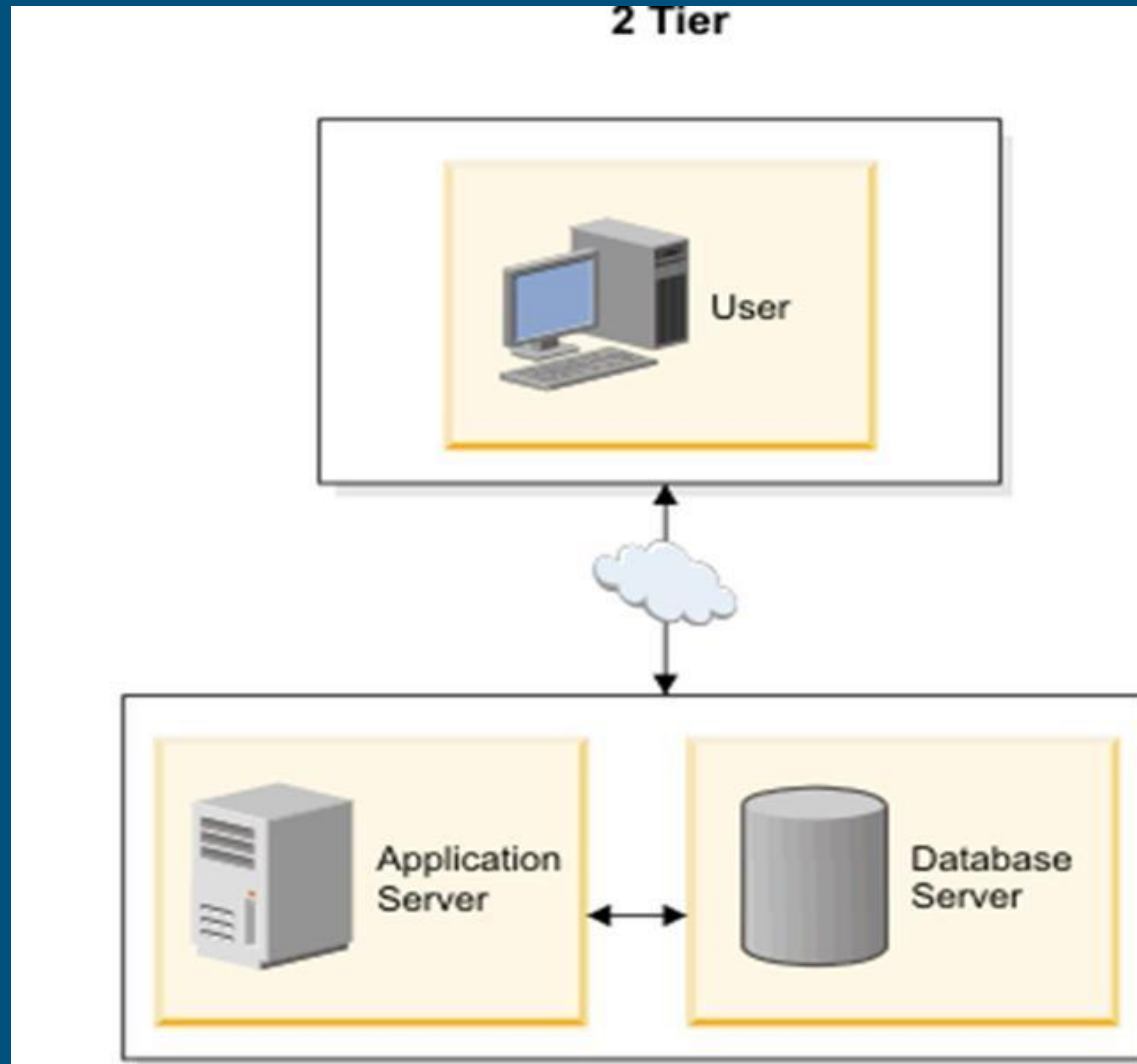
➤ Client tasks:

- Provide a way for users to submit queries
- Run applications that use the results of queries
- Display results of queries

➤ Server tasks:

- Implement a data manipulation language, which can directly access and update the database

System Architecture - – 2 tier



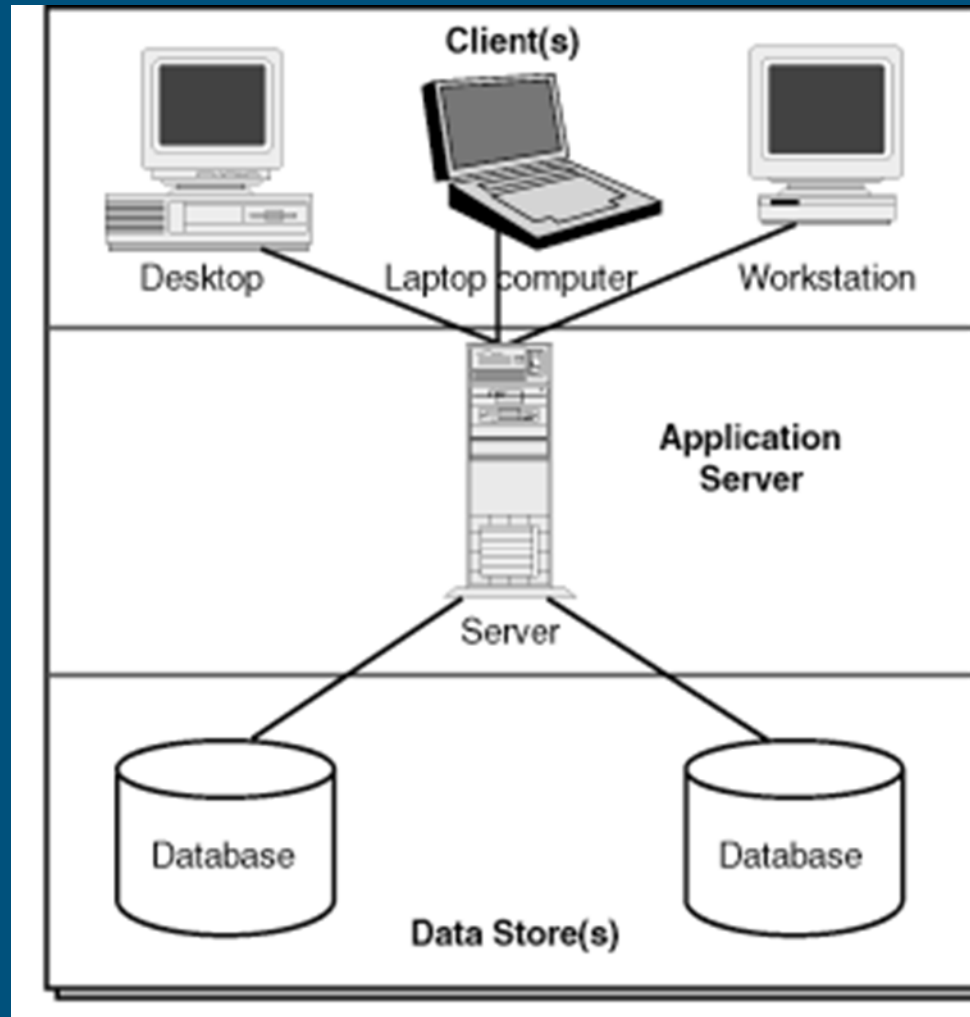
System Architecture - – 2 tier

- Problems with a two-tier system
 - Because the relative power of clients has grown considerably, we could shift processing to the client, but then maintaining data integrity is difficult

Solution

- Web server part of multi-tier application
 - Divide functionality into separate tiers
 - Logical groupings of functionality
 - Can reside on same computer or on different computers

System Architecture - – 3 tier



Client Tier

MiddleTier

InformationTier

System Architecture - – 3 tier

Information tier

- Referred to as the data tier or the bottom tier
 - Maintains data for the application
 - Stores data in a relational database management system
- ## Middle tier
- Implements business logic and presentation logic
 - Controls interactions between application clients and application data
 - Acts as an intermediary between data in the information tier and application clients

System Architecture - – 3 tier

- Middle tier
- Controller logic
 - Processes client requests from top tier
 - Retrieves data from database
- Presentation logic
 - Processes data from information tier
 - Presents content to client
- Business logic
 - Enforces business rules
 - Dictates how clients can access application data and how applications process data
 - Ensures data validity before updating the database

Core Communication Protocols



HTTP

Hypertext Transfer Protocol:
The foundation of data
exchange on the World Wide
Web.



HTTPS

Encrypted version of HTTP
using SSL/TLS to secure
sensitive user information.



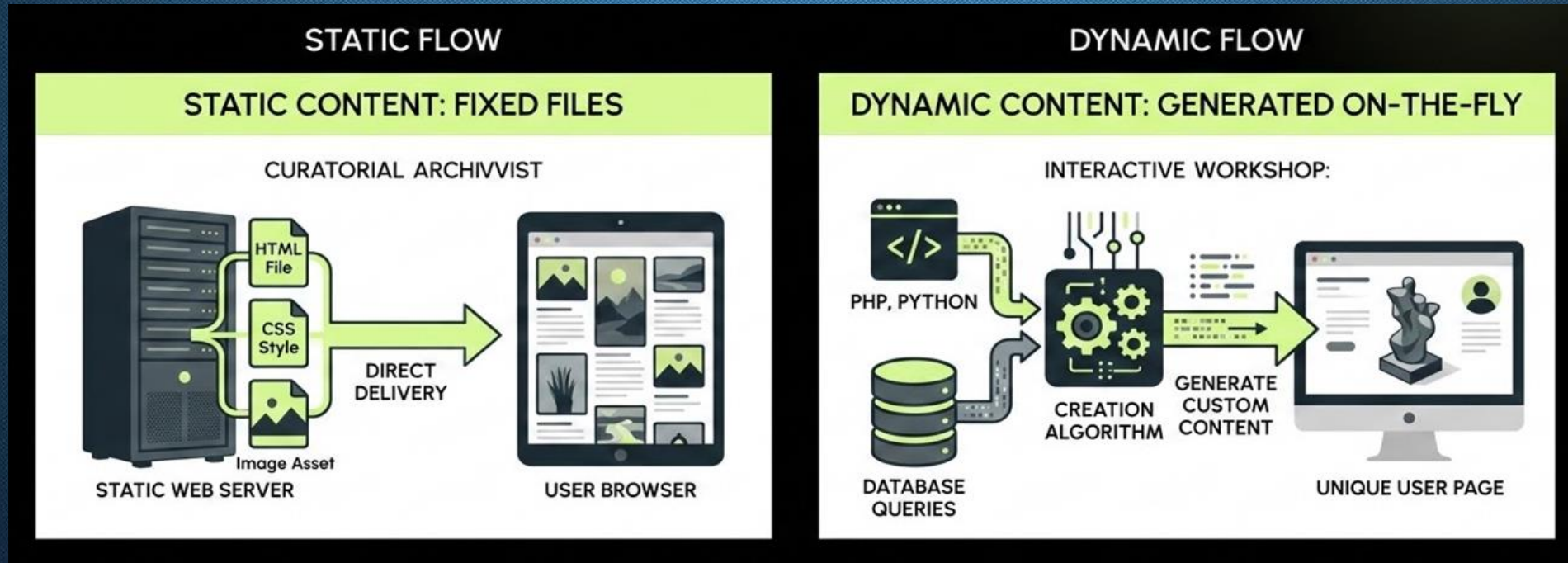
DNS

The internet's phonebook,
translating human-friendly
domains into IP addresses.

Parts of a Server



Types of Web Servers



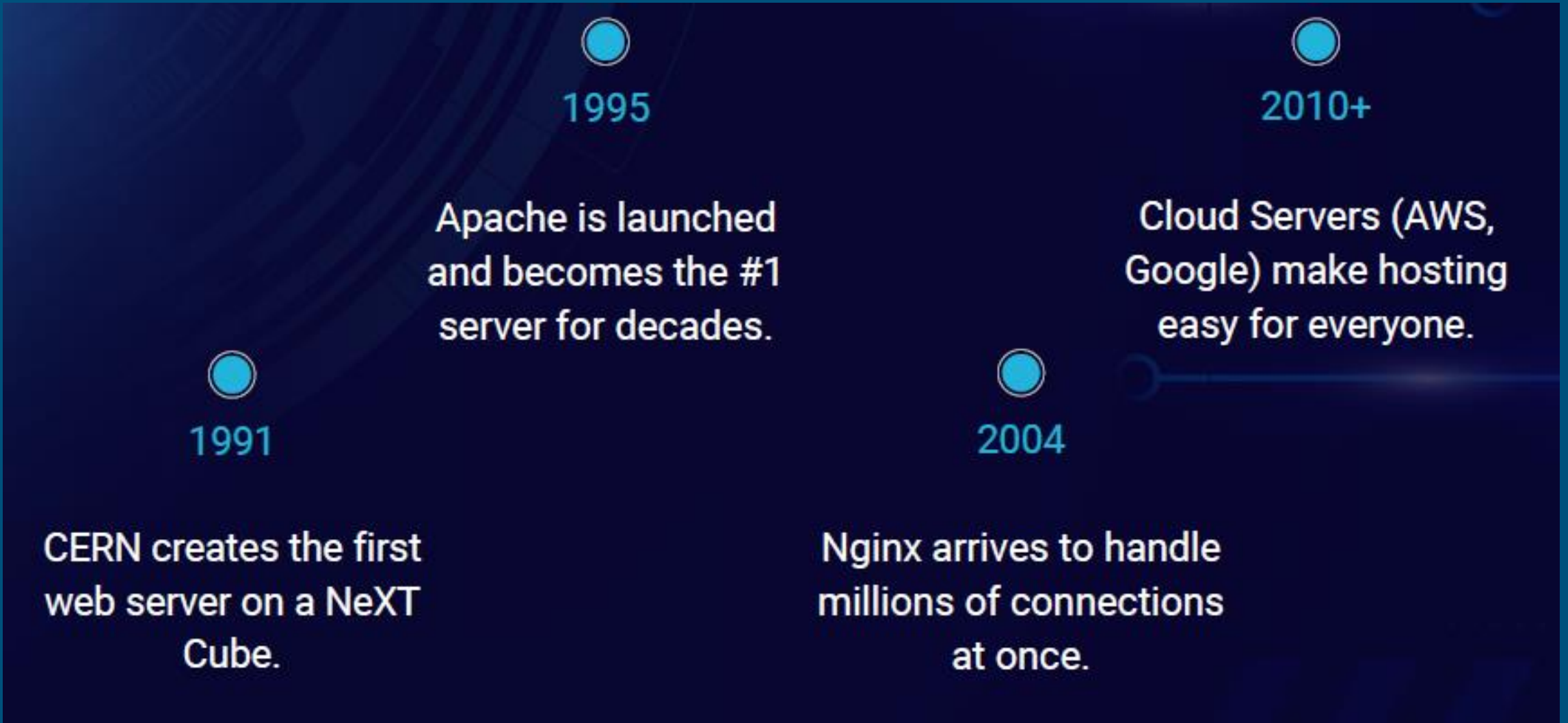
Static: Files served "as-is" (HTML, CSS, Images). High performance, easy to cache.

Dynamic: Server-side scripts (PHP, Python) generate custom content based on user data and database queries.

Static vs Dynamic Content

Feature	Static Content	Dynamic Content
Definition	Files sent exactly as stored	Generated on-the-fly by scripts
Speed	Lightning fast (low CPU)	Slower (requires processing)
Personalization	None (same for everyone)	High (unique per user)
Database	Not Required	Essential (MySQL, PostgreSQL)

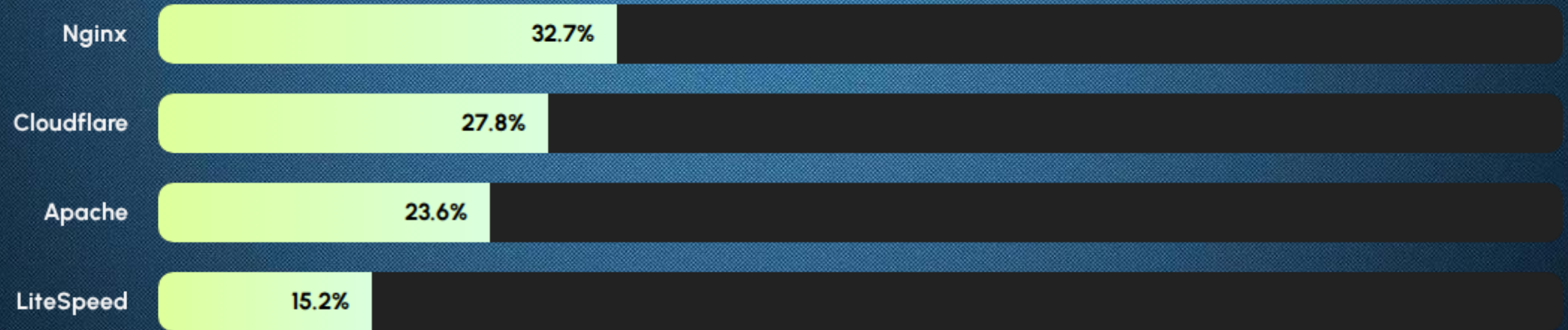
History of Web Servers



Industry Leading Server Software

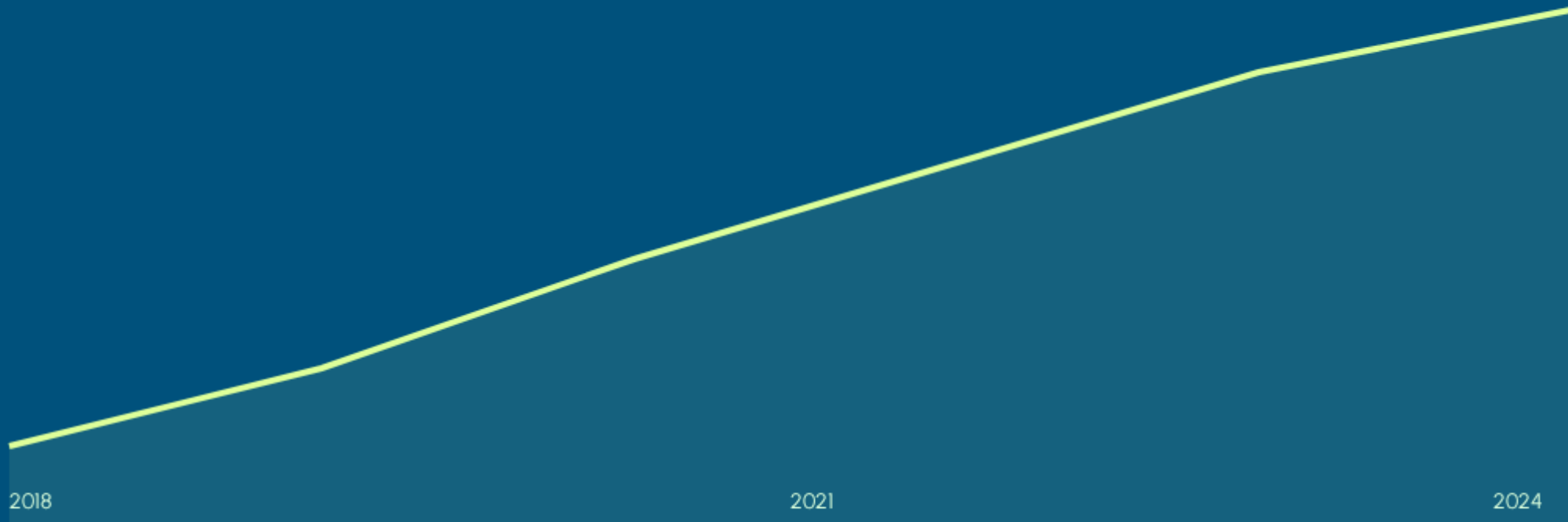
Server Software	Key Strength	Architecture
Apache	Extremely flexible & modular	Process-based
Nginx	High performance & scalability	Event-driven
Microsoft IIS	Native Windows/.NET integration	Thread-based
Caddy	Automatic HTTPS by default	Go-based

Global Market Share (2024)



Nginx's event-driven architecture allows it to handle significantly more simultaneous connections with a lower memory usage compared to traditional process-based servers.

Rise of Cloud & Edge Hosting



Trend: Shift from on-premise physical servers to distributed Cloud/Edge infrastructure

Why Do We Need Servers?

Always On: Websites must be available 24 hours a day, 365 days a year.

Security: Servers protect your website files from hackers and viruses.

Performance: High-end servers make sure your website loads fast for everyone.

Public Access: They give your local files a unique address (IP) so the world can find you.

Advantages of Web Servers

Easy access to websites worldwide

Fast sharing of information

Supports online services and applications

Can serve many users at the same time

Useful for businesses, education, and communication

Disadvantages of Web Servers

High setup and maintenance cost

Requires technical knowledge to manage

Security risks such as hacking and malware attacks

Server failures can make websites unavailable

Requires regular updates and monitoring

Security in Web Servers

Use HTTPS for secure communication.

Install firewalls and antivirus software.

Update server software regularly.

Use authentication and access control.

Backup important data frequently.

Critical Security Layers

SSL/TLS Certificates: Encrypts data in transit between client and server.

Authentication: Controlling access to resources via login credentials.

WAF & Firewalls: Protecting against DDoS, SQL injections, and malicious traffic.

Logging: Recording server activity to identify and troubleshoot security breaches.

For exploring the world of web servers

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